# Group Contribution Report

**Resource Reservation System**

## Project overview:

The campus resource reservation system isa group project created by Aashutosh Bishal and Eric using C++ language. This system allows students to view campus resources, create and cancel reservations, search for resources and reservations and join on waiting list. This projects uses various data structure components such as vector, linked list, stack, queues, linear search, merge sort. This report explains how team members worked on project and their contribution on each component.

## Team Contribution Summary:

| Name | Components | Task | Testing/Debugging | Github contribution |
|---|---|---|---|---|
| Aashutosh Khadka | Resource Management, File I/O Menu, | Resource class and resource data; load resources from file; file open validation; display resource ID, name, type, and availability; initial menu structure in main.cpp.<br>Also, required to review all files and test final project<br>Also, worked for Linear search, merge sort, resource utilization, final menu. | Verify valid file loading; test missing/failed file open; verify resource display and availability output; test menu input and integration points<br>Verify linear search, resource reports and menu integration | Created project file,<br>Add actual branch name, commits, and/or pull request |
| Eric | Reservation Management, Linked list | Reservation class and operations; linked list for active reservations; insertion, removal/cancellation, traversal, and display; reservation-request validation; duplicate reservation-ID prevention.<br>Worked on reservation searching, reservation start and | Test insertion and cancellation; test traversal/display; test duplicate IDs; test invalid requests and empty-list cases; debug linked-list pointer behavior<br>Verified reservation start and end time and reservation searching. | Add actual branch name, commits, and/or pull request |

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| | | end time and linked list utlilzation. | | |
| Bishal KC | Waiting Queue, Cancellation stacks, | FIFO waiting-list queue; add, remove, and display waiting students; LIFO cancellation-history stack; restore most recently cancelled reservation; display cancellation history | Test FIFO order; test empty waiting queue; test LIFO cancellation order; test undo/restore; test empty-stack handling and integration with cancellation flow. | Add actual branch name, commits, and/or pull request |

**Individual contribution:**

Aashutosh Khadka:

Aashutosh is responsible for the resource-management foundation of the system. His section covers the Resource class, resource information, and file input used to load the campus resource inventory. His work includes displaying each resource's ID, name, type, and availability status, handling basic file-opening errors, and creating the initial menu skeleton in main.cpp so the other team members' features can be integrated into one command-line program. Relevant files include Resource.h, Resource.cpp, resources.txt, and the initial main.cpp. Before submission, the team should confirm the specific tests, fixes, commits, branch names, and pull requests that reflect Aashutosh's actual work.

For the final project Aashutosh worked on final menu, resource report and linear search and merge sort. He implemented linear search and merge sort in resource.cpp, worked on resource reports in main menu and added utilization reporting. At last Aashutosh combined all files together and did testing and debugging and make submission.

Eric:

Eric is responsible for reservation management and the linked list that stores active reservations. His section includes creating and managing reservation data, inserting reservations, removing/cancelling reservations, traversing the linked list, and displaying active reservation information. His responsibilities also include validating reservation requests and helping prevent duplicate reservation IDs. Relevant files include Reservation.h, Reservation.cpp, ReservationMain.h, and ReservationMain.cpp. Before submission, the team should confirm the specific tests, fixes, commits, branch names, and pull requests that reflect Eric's actual work.

Eric worked on searching reservation ID, also he added reservation start time and end time. He integrate active reservation with other component.

Bishal:

Bishal is responsible for the waiting-list queue and cancellation-history stack. His section implements FIFO behavior for students waiting on unavailable resources, including adding, removing, and displaying students in the queue. He also implements LIFO cancellation history, including storing cancelled reservations, restoring the most recently cancelled reservation, and displaying cancellation history. Relevant files include WaitingList.h/.cpp and CancellationHistory.h/.cpp. Before submission, the team should confirm the specific tests, fixes, commits, branch names, and pull requests that reflect Bishal actual work

Bishal worked on connecting waiting list to reservation cancellation. He assigned a resource to next waiting student and he tested FIFO queue and LIFO stack. He debugged waiting list and undo operation.

**Shared Team Responsibilities:**

Integrate all three sections into one working program and test the combined system. Complete the README and complexity analysis together. Use one shared GitHub repository, with meaningful contributions from each member's own account. Create and merge at least one feature branch and review the combined code before merging to main. Verify that the final submitted ZIP matches the source and header files in the GitHub repository and that the project compiles and runs on UNT CSE CELL machines.